Data Communication

CSE 2211

Chapter 6 : Multiplexing-FDM

6-1 MULTIPLEXING

- ***Multiplexing is the set of techniques that allows the (simultaneous) transmission of multiple signals across a single data link.***
  - *The word **link** refers to the physical path.*
  - *The word channel refers to the portion of a link that carries a transmission between a given pair of lines.*
  - *One link can have many (n) channels.*

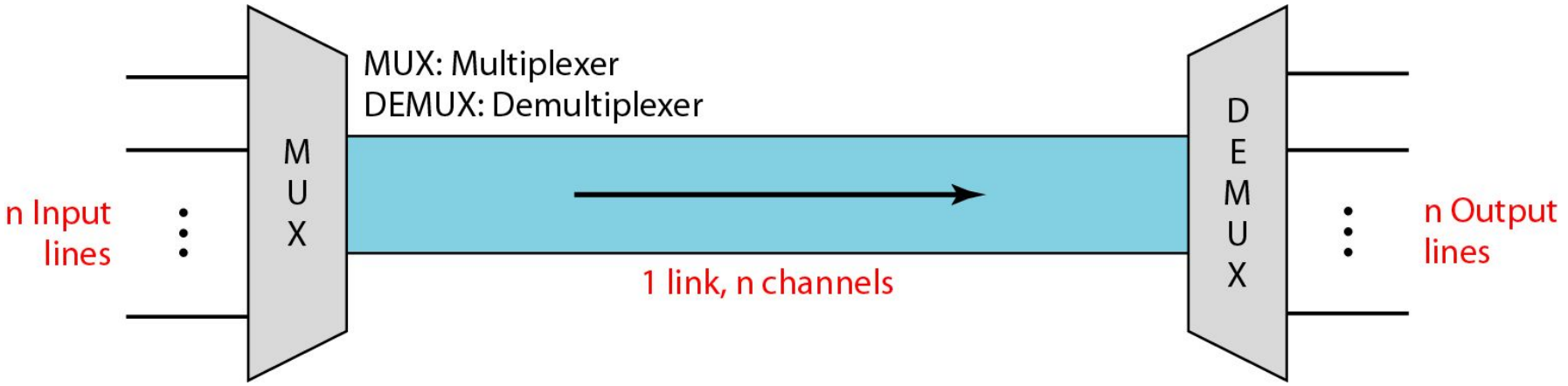


Figure 6.1 *Dividing a link into channels*

- *In a multiplexed system, n lines share the bandwidth of one link. Figure below shows the basic format of a multiplexed system.*

- *The lines on the left direct their transmission streams to a multiplexer (MUX), which combines them into a single stream (many-to one).*

- *At the receiving end, that stream is fed into a demultiplexer (DEMUX), which separates the stream back into its component transmissions (one-to-many) and directs them to their corresponding lines.*

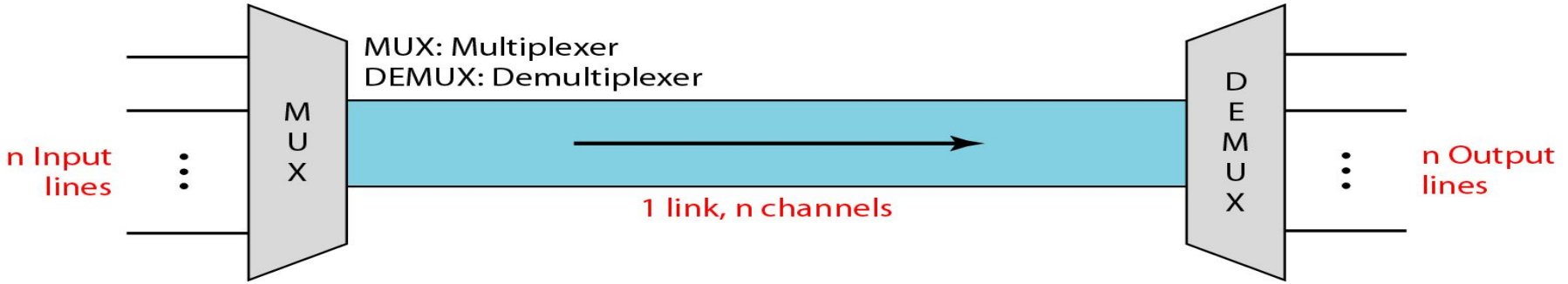
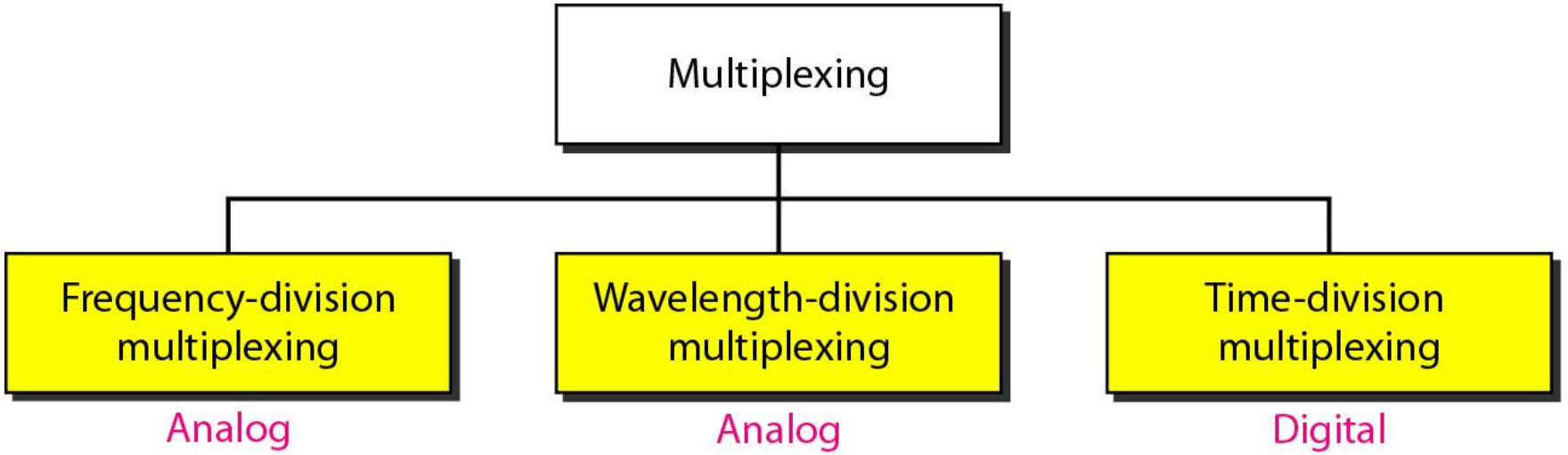


Figure 6.2 *Categories of multiplexing*

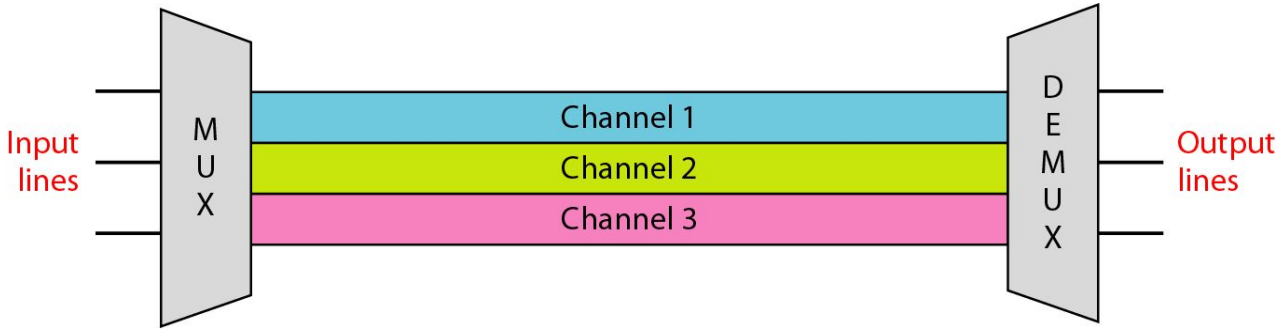


Frequency-division multiplexing

- *Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.*

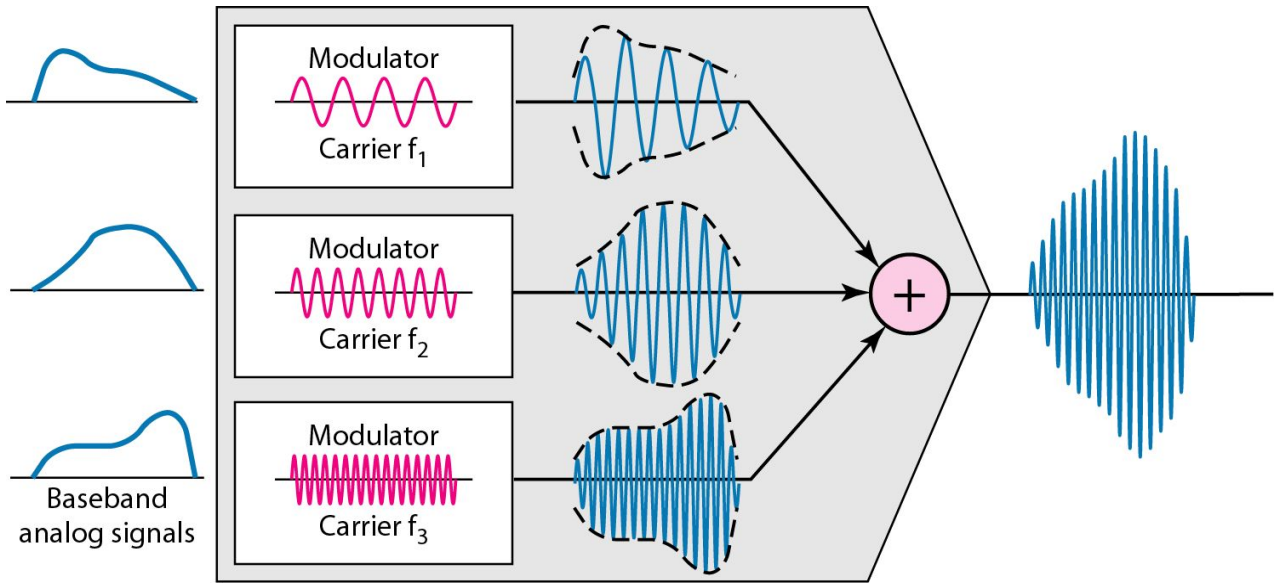
- *In FDM, signals generated by each sending device modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link.*

- *Channels can be separated by strips of unused bandwidth-guard bands-to prevent signals from overlapping.*



Multiplexing Process

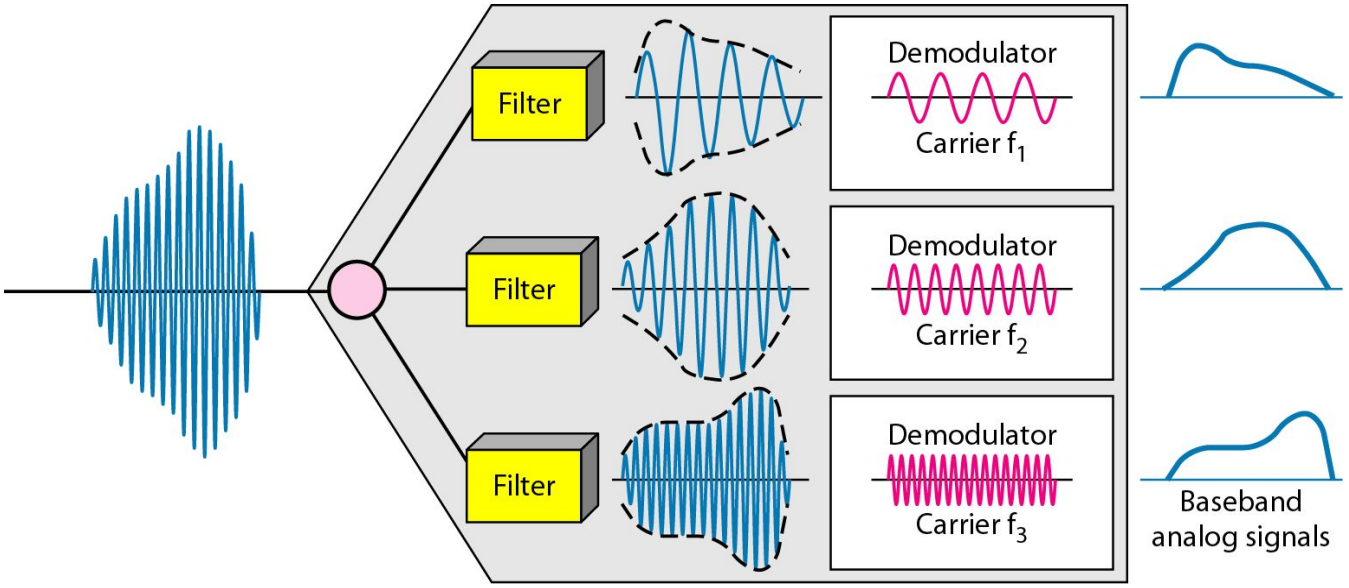
- *Each source generates a signal of a similar frequency range.*
- *Inside the multiplexer, these similar signals modulate different carrier frequencies (f_1, f_2 and f_3).*
- *The resulting modulated signals are then combined into a single composite signal that is sent out over a media link that has enough bandwidth to accommodate it.*



Demultiplexing Process

The demultiplexer uses a series of filters to decompose the multiplexed signal into its constituent component signals.

The individual signals are then passed to a demodulator that separates them from their carriers and passes them to the output lines.



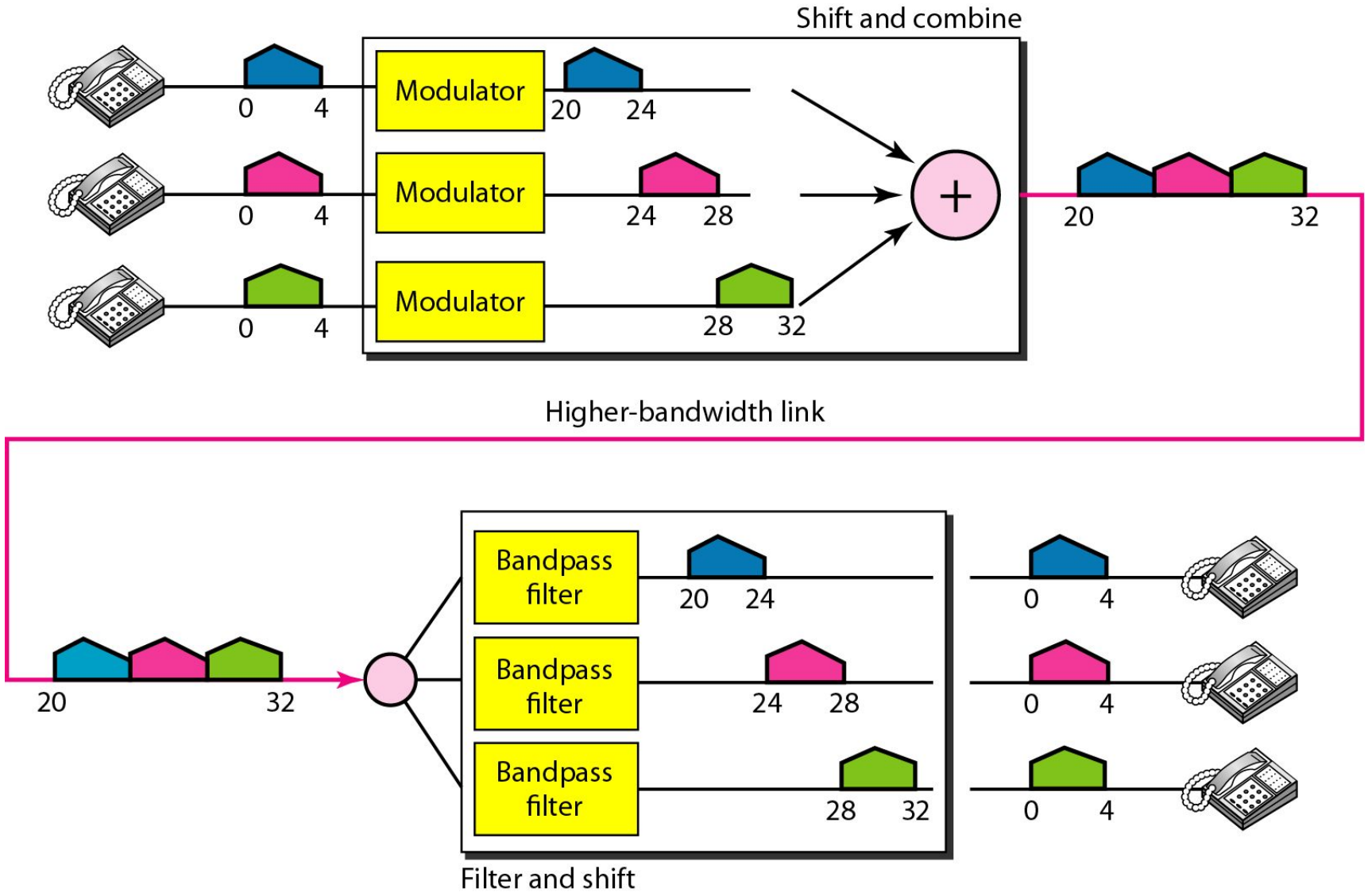
Example 6.1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 6.6. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 6.6.

Figure 6.6 *Example 6.1*



Example 6.2

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

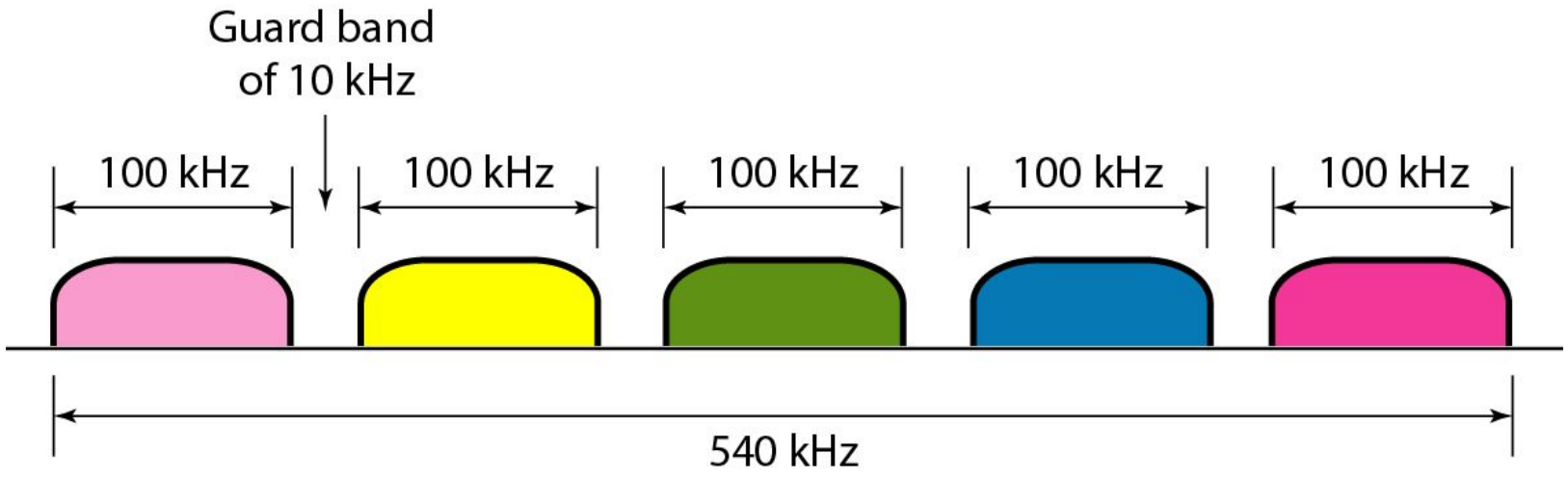
Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$

as shown in Figure 6.7.

Figure 6.7 *Example 6.2*



Example 6.3

Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM.

Solution

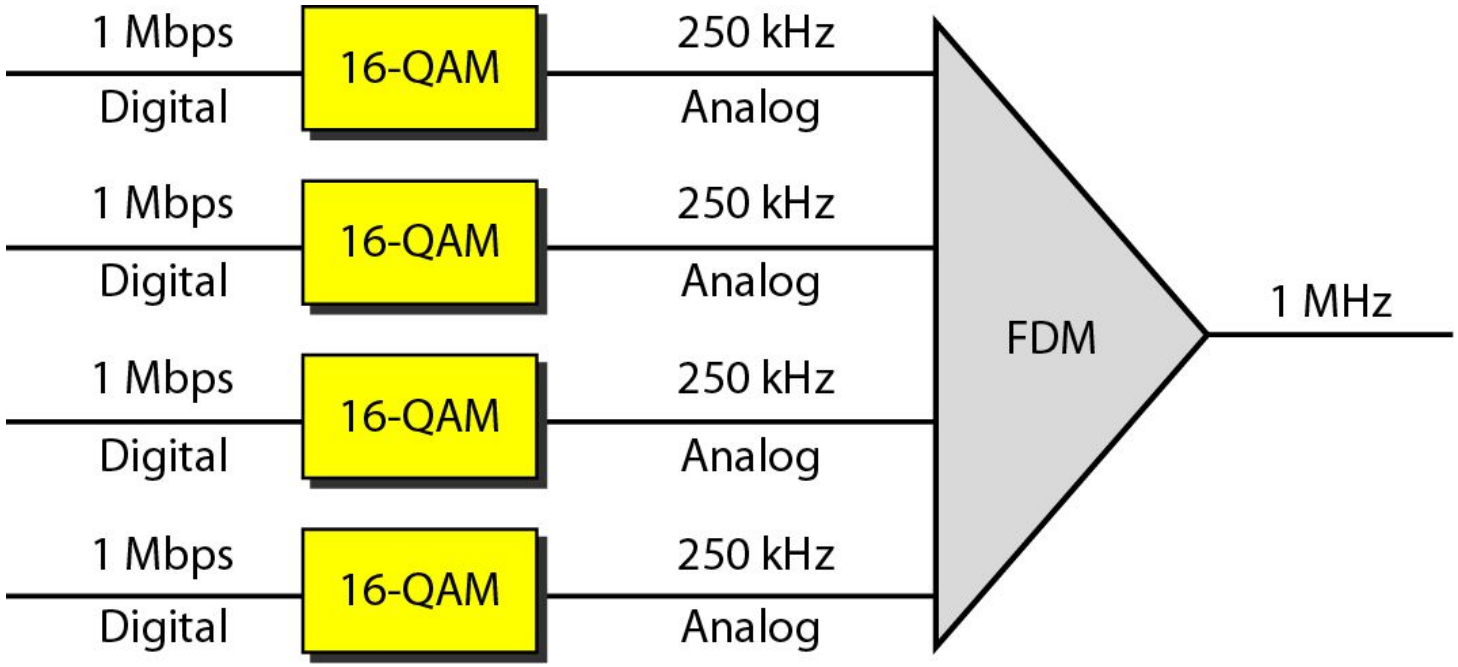
The satellite channel is analog. We divide it into four channels, each channel having 1M/4=250-kHz bandwidth.

Each digital channel of 1 Mbps must be transmitted over a 250KHz channel.

Bandwidth of QAM B=(1+d)S); Here, d=0.

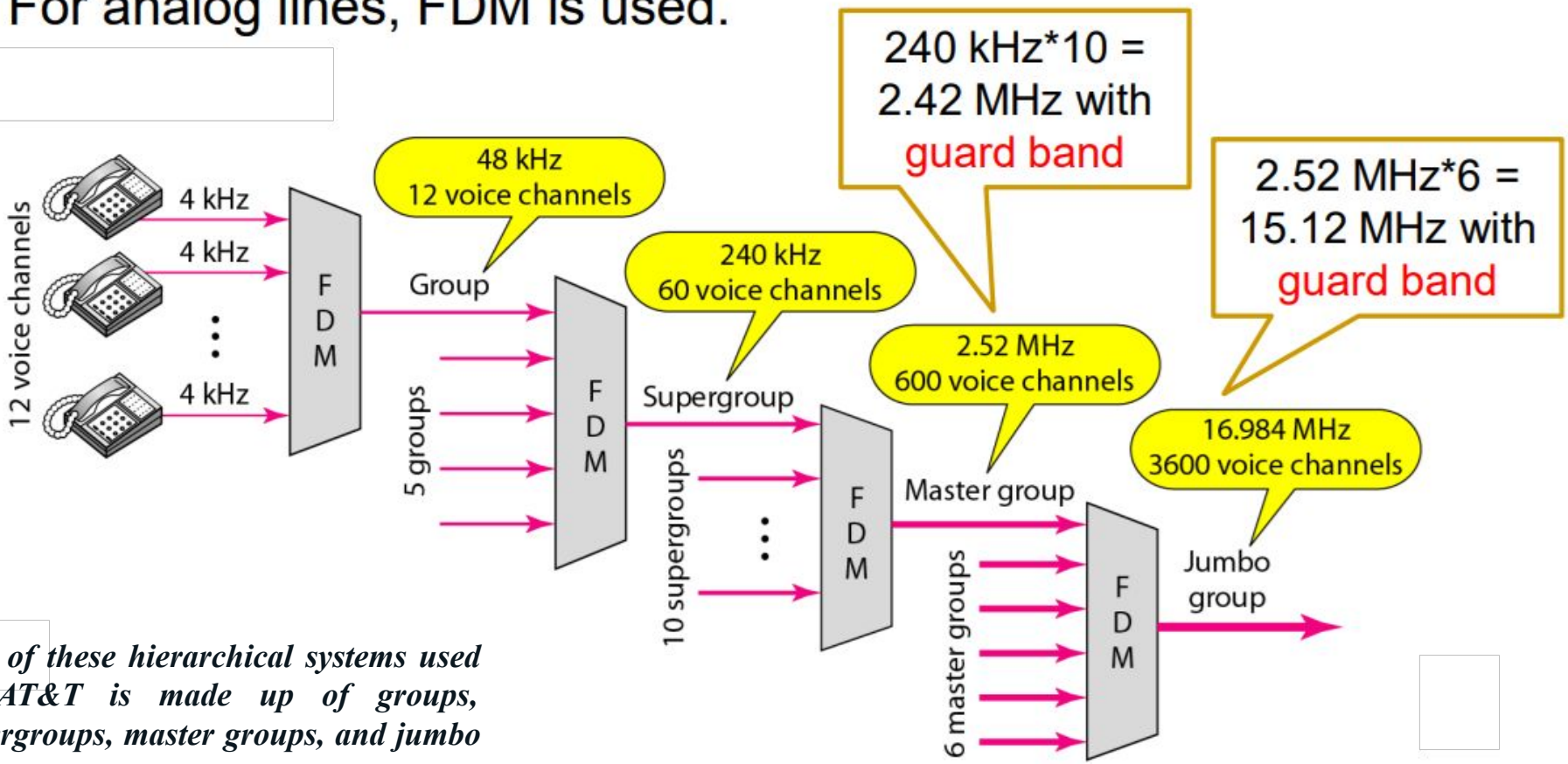
r = 4 bits/signal elements
One solution is 16-QAM modulation.

Figure 6.8 *Example 6.3*



Analog hierarchy

- To maximize the efficiency of infrastructure, we can multiplex signals from lower-bandwidth lines to higher-bandwidth lines.
- Many leased lines can be combined into fewer but bigger channels.
- For analog lines, FDM is used.



One of these hierarchical systems used by AT&T is made up of groups, supergroups, master groups, and jumbo groups.

Applications of FDM-AM and FM radio broadcasting

A very common application of FDM is AM and FM radio broadcasting.
Radio uses the air as the transmission medium.
A special band from 530 to 1700 kHz is assigned to AM radio.
All radio stations need to share this band.
Each AM station needs 10 kHz of bandwidth.
Each station uses a different carrier frequency, which means it is shifting its signal and multiplexing.
The signal that goes to the air is a combination of signals.
A receiver receives all these signals, but filters (by tuning) only the one which is desired.

AM and FM radio broadcasting-Cont.

Without multiplexing, only one AM station could broadcast to the common link, the air.
However, we need to know that there is physical multiplexer or demultiplexer here.
The situation is similar in FM broadcasting. However, FM has a wider band of 88 to 108 MHz because each station needs a bandwidth of 200 kHz.

Television broadcasting

Another common use of FDM is in television broadcasting. Each TV channel has its own bandwidth of 6 MHz.

Cellular telephone

- *The systems divide the bandwidth that is available to them into multiple channels.*
- *Thus, the telephone connection of one user is assigned one set of frequencies for transmission, while, the telephone connection of a second user is assigned a second set of frequencies.*
- *First generation cellular telephone systems allocated channels using frequency ranges within the 800-to-900 MHz spectrum.*
- *To be more precise, the 824-to-849-MHz range was used for receiving signals from cellular phones (the uplink), while the 869-to-894-MHz range was used for transmitting to cellular phone (the downlink).*

Cont.

- *To enable a two-way conversation to occur, two channels were assigned to each telephone connection.*
- *Each user is assigned two 30-kHz channels, one for sending voice and the other for receiving.*
- *The voice signal, which has a bandwidth of 3 kHz (from 300 to 3300 Hz), is modulated by using FM.*
- *Remember that an FM signal has a bandwidth 10 times that of the modulating signal, which means each channel, has 30 kHz (10 x 3) of bandwidth.*
- *Therefore, each user is given, by the base station, a 60-kHz bandwidth in a range available at the time of the call.*

Example 6.4

The Advanced Mobile Phone System (AMPS) uses two bands. The first band of 824 to 849 MHz is used for sending, and 869 to 894 MHz is used for receiving. Each user has a bandwidth of 30 kHz in each direction. How many people can use their cellular phones simultaneously?

Solution

Each band is 25 MHz. If we divide 25 MHz by 30 kHz, we get 833.33. In reality, the band is divided into 832 channels. Of these, 42 channels are used for control, which means only 790 channels are available for cellular phone users.

Note

We need to know that there is no physical multiplexer or demultiplexer here. Multiplexing is done at the data-link layer.